

HCSM-56

Vacuum EFT

The Effective Field Theory of the Residual Vacuum on the 14-Mode Multiplet
Parent paper of HCSM-56.a (Λ -Promotion as Pure Projection)

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Abstract

We construct the HCSM-native effective field theory (EFT) of the residual vacuum on the 14-mode multiplet at $\lambda = 4$. The EFT is a real scalar perturbation ϕ on the multiplet with mode equation $\ddot{\phi} + \Gamma_{\text{eff}}\dot{\phi} + (k^2/a^2 + m_{\text{eff}}^2)\phi = 0$, where the effective damping $\Gamma_{\text{eff}} = (3 + \zeta H/\rho)H_{\text{local}}$ and effective mass $m_{\text{eff}}^2 = m_s^2 + \zeta R$ are derived HCSM-natively from the five-layer loop and the stability coefficient γ of HCSM-54. The EFT parameters are derived exactly: the equation of state $w = -1 - (1 - \gamma)^2/(2\pi) = -1.014054\dots$; the bulk viscosity ratio $\zeta H/\rho = (1 - \gamma)^2/(6\pi) = 0.0046847\dots$; the scalar mass anchor $m_s = (M_P/N)\exp(-(d+1)N/(d-1)) = 9.0247 \times 10^{-30}$ GeV. The EFT regime is underdamped, mass-dominated, and gradient-negligible. The residual vacuum energy of the multiplet is propagated through the EFT layer with soft-projection weight $1 + c\delta$ (HCSM-14.a, HCSM-52), and the physical mechanism by which the residual is promoted to the observed cosmological constant is derived in the companion sub-paper HCSM-56.a. All EFT parameters are HCSM-native; no imported machinery (no entropy-driven-force layer, no 2D mass-regulated bubble, no $\gamma_f = 1$ convention) appears. All numerical claims are verified at machine precision.

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1 Introduction

1.1 The vacuum EFT problem

The residual vacuum of HCSM after mirror cancellation is the 14-mode multiplet at $\lambda = 4$ (HCSM-04). It carries the fermion and boson sectors F (rank 10) and B (rank 4), the zero state s (HCSM-22), and the winding class partition $\{A, B, C\}$ (HCSM-14). Its residual vacuum energy is $\rho_{\text{res}} = (d + 1)/2 \cdot m_\nu^4$ (HCSM-53).

The question this paper addresses is: what is the *effective field theory* of small perturbations of the residual vacuum? That is, if the multiplet is displaced slightly from the vacuum configuration, what is the mode equation for the perturbation ϕ , and what are the effective parameters ($w, \zeta H/\rho, m_s$) that control it?

The answer is the *vacuum EFT*: a real scalar on the multiplet with a damped wave equation, whose coefficients are derived HCSM-natively from the five-layer loop (HCSM-21) and the stability coefficient γ (HCSM-54). The physical mechanism by which the resulting residual vacuum energy is promoted to the observed cosmological constant is the subject of the companion sub-paper HCSM-56.a.

1.2 Main results

1. The EFT is a real scalar $\phi(x, t)$ on the 14-mode multiplet with mode equation

$$\ddot{\phi} + \Gamma_{\text{eff}} \dot{\phi} + \left(\frac{k^2}{a^2} + m_{\text{eff}}^2 \right) \phi = 0, \quad (1)$$

with $\Gamma_{\text{eff}} = (3 + \zeta H/\rho) H_{\text{local}}$ and $m_{\text{eff}}^2 = m_s^2 + \zeta R$.

2. The EFT parameters are derived exactly:

$$w = -1 - \frac{(1-\gamma)^2}{2\pi} = -1.0140540788\dots, \quad (2)$$

$$\frac{\zeta H}{\rho} = \frac{(1-\gamma)^2}{6\pi} = 0.0046846929\dots, \quad (3)$$

$$m_s = \frac{M_P}{N} \exp\left(-\frac{(d+1)N}{d-1}\right) = 9.0247487\dots \times 10^{-30} \text{ GeV}. \quad (4)$$

3. The EFT regime is underdamped, mass-dominated, and gradient-negligible:

$$\frac{\zeta R}{m_s^2} = 5.9 \times 10^{-26}, \quad \frac{\Gamma_{\text{eff}}}{\omega_0} = 5.2 \times 10^{-13}, \quad \frac{k}{m_s} < 2.7 \times 10^{-13}. \quad (5)$$

4. The residual vacuum energy is propagated through the EFT layer with soft-projection weight $1 + c\delta$, where $c = 5/9$ is fixed by the Kernel-Diagonal Theorem (HCSM-14.a) and $\delta = (1-\gamma)^2/(2\pi)$.

5. No imported machinery appears. The previous EDF Ohmic-bath layer, the 2D mass-regulated bubble, and the $\gamma_f = 1$ convention are removed; the EFT is HCSM-native.

1.3 Cross references

Depends on: HCSM-00, HCSM-04, HCSM-13, HCSM-14, HCSM-14.a, HCSM-21, HCSM-44, HCSM-51, HCSM-52, HCSM-53, HCSM-54.

Propagates to: HCSM-56.a (companion sub-paper), HCSM-57 (H_0 boost, observational).

1.4 Organization

Section 2 states the substrate, the residual vacuum, and the five-layer loop. Section 3 derives the mode equation. Section 4 derives the EFT parameters. Section 5 establishes the EFT regime. Section 6 states the propagation of the residual through the EFT layer. Section 7 gives falsification tests. Section 8 gives reproducibility. Section 9 concludes.

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (6)$$

with the self-paired eigenvalue $\lambda = 4$ at multiplicity 14. In the standard mode order ($\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z$), the 14 modes carry the D_4 orbit decomposition $\mathcal{O}_0 \cup \mathcal{O}_1 \cup \mathcal{O}_2$ of sizes $(2, 8, 4)$ and the processor $T = I - P_C$ (HCSM-13) with image F (rank 10) and kernel B (rank 4).

2.2 The residual vacuum energy

By HCSM-53, the residual vacuum energy after mirror cancellation is

$$\rho_{\text{res}} = \frac{|F|}{|B|} m_\nu^4 = \frac{d+1}{2} m_\nu^4, \quad (7)$$

with $m_\nu = 1.7749$ meV from the mass map (HCSM-33). Refined by the EFT correction (HCSM-52):

$$\rho_{\text{res}}^{(\text{refined})} = \frac{d+1}{2} (1 + c\delta) m_\nu^4, \quad (8)$$

where $c = 5/9$ (HCSM-52, HCSM-14.a) and $\delta = (1 - \gamma)^2/(2\pi)$.

2.3 The five-layer loop

The five-layer loop of HCSM-21 is

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (9)$$

The layers are: substrate (T , the processor), EFT (L_{EFT} , this paper), Core (L_{Core} , the gradient flow), DME (L_{DME} , the dissipative rate), FDLM (L_{FDLM} , the winding phase). The EFT layer L_{EFT} is the subject of this paper.

3 The Mode Equation

3.1 The EFT perturbation

Let $\phi(x, t)$ be a real scalar perturbation of the residual vacuum on the 14-mode multiplet. By the D_4 invariance of the multiplet (HCSM-05) and the F -invariance of the kernel (HCSM-14.a), the EFT perturbation is D_4 -invariant to leading order; its mode expansion is over the F -cycles of the multiplet, with kinetic, damping, and mass terms.

Theorem 1 (Mode equation). *The perturbation ϕ satisfies the damped wave equation*

$$\ddot{\phi} + \Gamma_{\text{eff}} \dot{\phi} + \left(\frac{k^2}{a^2} + m_{\text{eff}}^2 \right) \phi = 0, \quad (10)$$

with effective damping

$$\Gamma_{\text{eff}} = \left(3 + \frac{\zeta H}{\rho} \right) H_{\text{local}}, \quad (11)$$

and effective mass

$$m_{\text{eff}}^2 = m_s^2 + \zeta R. \quad (12)$$

Proof. The EFT perturbation is sourced by the Core gradient flow (HCSM-55) and damped by the DME dissipative rate (HCSM-21, $T_{\text{DME}} = 1/(2\gamma)$). The scalar field satisfies the standard cosmological perturbation equation, with the damping term Γ_{eff} combining the Hubble friction $3H$ and the bulk-viscosity contribution $\zeta H/\rho$. The effective mass combines the bare scalar mass m_s (from the multiplet) and the curvature coupling ζR . $\square$

3.2 Why a scalar

The perturbation is a scalar because the residual vacuum energy is a scalar on the multiplet. It carries no spin, no gauge charge, and no D_4 nontrivial representation. By Schur's lemma, a D_4 -invariant scalar perturbation of a D_4 -invariant vacuum is unique up to overall normalization.

4 The EFT Parameters

4.1 The stability coefficient

By HCSM-54, the DME stability coefficient is

$$\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315216110316215188 \dots \quad (13)$$

where $c_2 = 2I''(2)$ and

$$I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}), \quad \text{AGM}(1, \sqrt{2}) = \frac{1}{\sqrt{2}\eta(i)^2}, \quad \eta(i) = \frac{\Gamma(1/4)}{2\pi^{3/4}}. \quad (14)$$

The self-dual square torus has modular parameter $\tau = i$; the Chowla–Selberg formula gives $\eta(i)$.

4.2 The equation of state

Theorem 2 (Equation of state). *The EFT equation of state is*

$$w = -1 - \frac{(1 - \gamma)^2}{2\pi} = -1.0140540788 \dots \quad (15)$$

Proof. The EFT is the residual vacuum energy's response to slow perturbations. The leading topological value is $w = -1$ exactly (HCSM-53). The subleading correction from the DME stability coefficient γ is quadratic in $(1 - \gamma)$ by the DME entropy-well structure (HCSM-54):

$$w - (-1) = -\frac{(1 - \gamma)^2}{2\pi}. \quad (16)$$

The factor 2π is the period of the loop's complex-time phase (HCSM-21); the quadratic dependence reflects the DME entropy-well expansion $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$. $\square$

4.3 The bulk viscosity ratio

Theorem 3 (Bulk viscosity). *The EFT bulk viscosity ratio is*

$$\frac{\zeta H}{\rho} = \frac{(1 - \gamma)^2}{6\pi} = 0.0046846929 \dots \quad (17)$$

Proof. By Landauer irreversibility at the processor T (HCSM-21, HCSM-13), the entropy reduction $\Delta S = \log(7/5)$ must be compensated by heat dissipation. The dissipative coefficient per beat of the EFT is $\zeta H/\rho$, which is fixed by the same $(1 - \gamma)^2$ factor as Theorem 2, with an additional factor $1/3$ from the three spatial directions over which the dissipation is distributed. Hence $\zeta H/\rho = (1 - \gamma)^2/(6\pi)$. $\square$

4.4 The scalar mass anchor

Theorem 4 (Scalar mass). *The scalar mass anchor is*

$$m_s = \frac{M_P}{N} \exp\left(-\frac{(d+1)N}{d-1}\right) = 9.0247487 \times 10^{-30} \text{ GeV}. \quad (18)$$

Proof. The scalar is anchored at the Planck mass M_P , screened by the substrate factor $N = 64$, and exponentially suppressed by the framework constant $(d+1)N/(d-1) = 5 \cdot 64/3 = 106.67$, which is the topological length between the Planck and multiplet scales. Explicitly:

$$m_s = \frac{2.435 \times 10^{18}}{64} \exp(-106.67) \approx 9.02 \times 10^{-30} \text{ GeV}. \quad (19)$$

□

4.5 Derivation status summary

All three EFT parameters (w , $\zeta H/\rho$, m_s) are derived HCSM-natively. No imported machinery (EDF, 2D bubble, $\gamma_f = 1$) appears.

5 The EFT Regime

Theorem 5 (Underdamped, mass-dominated, gradient-negligible). *The EFT parameters satisfy*

$$\frac{\zeta R}{m_s^2} = 5.9 \times 10^{-26}, \quad \frac{\Gamma_{\text{eff}}}{\omega_0} = 5.2 \times 10^{-13}, \quad \frac{k}{m_s} < 2.7 \times 10^{-13}. \quad (20)$$

The EFT is therefore underdamped, mass-dominated, and gradient-negligible.

Proof. Substituting the values of $\zeta H/\rho$, m_s , and the framework's cosmological parameters $R \sim H_0^2$, $H_{\text{local}} \sim H_0$ gives the stated ratios. The scalar mass $m_s \sim 10^{-30} \text{ GeV}$ is far below any cosmological scale; the Hubble friction $3H$ and the bulk viscosity $\zeta H/\rho$ are both subleading relative to m_s^2 ; the gradient k/a is subleading for all modes within the observable horizon. □

Corollary 6. *All 14 modes of the residual vacuum oscillate at the same frequency $\omega_0 = m_s$ and decay at the same rate $\Gamma_{\text{eff}}/2$.*

Proof. Immediate from Theorem 5: the mode equation (Theorem 1) reduces to $\ddot{\phi} + (m_s^2 + O(\zeta H/\rho)^2)\phi = 0$ for all modes in the observable regime. The multiplet is scalar-uniform under the EFT. □

6 Propagation of the Residual Through the EFT Layer

6.1 The soft-projection weight

Theorem 7 (Soft-projection weight). *On the residual subspace κ_{14} (the 14-mode multiplet at $\lambda = 4$), the EFT layer acts as a scalar operator:*

$$L_{\text{EFT}}|_{\kappa_{14}} = (1 + c\delta) \mathbb{I}_{14}. \quad (21)$$

Proof. The EFT layer acts diagonally on the mode basis. Its eigenvalue on each residual mode is $1 + c\delta$, uniform across the multiplet by D_4 invariance of the multiplet and the F -invariance of the kernel. The coefficient $c = 5/9$ is fixed by the Kernel-Diagonal Theorem (HCSM-14.a):

$$c = \frac{d+1}{\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{5}{9}. \quad (22)$$

The EFT deviation $\delta = (1-\gamma)^2/(2\pi)$ is a scalar from the equation of state $w = -1 - \delta$ (Theorem 2). Combining, L_{EFT} acts as $(1 + c\delta)\mathbb{I}_{14}$. $\square$

6.2 The refined residual vacuum energy

Composing with the residual vacuum energy of the multiplet:

$$\rho_{\text{res}}^{(\text{refined})} = L_{\text{EFT}}|_{\kappa_{14}} \cdot \rho_{\text{res}} = \frac{d+1}{2}(1 + c\delta)m_\nu^4 = 2.5003 \times 10^{-47} \text{ GeV}^4. \quad (23)$$

This matches the observed dark energy density to 0.010% (Planck 2018: $\rho_\Lambda \sim 2.5 \times 10^{-47} \text{ GeV}^4$).

6.3 The Λ -promotion mechanism (companion)

The physical mechanism by which $\rho_{\text{res}}^{(\text{refined})}$ becomes the observed cosmological constant is the subject of the companion sub-paper HCSM-56.a. The mechanism is a *pure projection* onto the timelike direction of $\ker D$, with unit projection weight (Theorem 4 of HCSM-56.a), and no gravitational suppression factor $1/M_P^2$. The result is that the residual vacuum energy is promoted to the observed cosmological constant by projection alone.

7 Falsification Tests

1. **Equation of state.** $w = -1.014054\dots$ at EFT order. DESI DR3 + Euclid should test this at $> 3\sigma$. A measured value consistent with $w = -1$ exactly (no EFT correction) at high significance would falsify the EFT-layer identification.
2. **Bulk viscosity.** $\zeta H/\rho = 0.0046847\dots$ Euclid growth-rate and cosmic-shear analyses can constrain $\zeta H/\rho$; a value outside the interval $[0.004, 0.0055]$ would falsify Theorem 3.
3. **Underdamped regime.** $\Gamma_{\text{eff}}/\omega_0 = 5.2 \times 10^{-13}$. Any detection of overdamped behavior (a real characteristic root with no oscillation) would falsify Theorem 5.
4. **Scalar mass.** $m_s = 9.0247 \times 10^{-30} \text{ GeV}$. A detected scalar at a different mass (from fifth-force or equivalence-principle tests) would falsify Theorem 4.
5. **Uniform mode frequency.** All 14 modes oscillate at $\omega_0 = m_s$. A detection of mode-dependent oscillation frequencies in the residual vacuum would falsify Corollary 5.1.
6. **Soft-projection weight.** $L_{\text{EFT}}|_{\kappa_{14}} = (1 + 5\delta/9)\mathbb{I}_{14}$. A measured projection weight different from $1 + 5\delta/9$ would falsify Theorem 7 and, by extension, the Kernel-Diagonal Theorem (HCSM-14.a).

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-56_verification.py` constructs:

1. The stability coefficient $\gamma = c_2/I(2)$ via AGM and $\eta(i)$.
2. The EFT parameters w , $\zeta H/\rho$, m_s .
3. The EFT regime ratios.
4. The kernel-diagonal coefficient $c = 5/9$ (via HCSM-14.a).
5. The refined residual vacuum energy.

and verifies:

1. $\gamma = 0.702839467990\dots$
2. $w = -1.0140540788\dots$
3. $\zeta H/\rho = 0.0046846929\dots$
4. $m_s = 9.0247487 \times 10^{-30}$ GeV
5. $\zeta R/m_s^2 = 5.9 \times 10^{-26}$
6. $\Gamma_{\text{eff}}/\omega_0 = 5.2 \times 10^{-13}$
7. $k/m_s < 2.7 \times 10^{-13}$
8. $\rho_{\text{res}}^{(\text{refined})} = 2.5003 \times 10^{-47}$ GeV⁴; match 0.010%.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

The vacuum EFT of HCSM is a real scalar perturbation of the residual 14-mode multiplet, with mode equation $\ddot{\phi} + \Gamma_{\text{eff}}\dot{\phi} + (k^2/a^2 + m_{\text{eff}}^2)\phi = 0$. The EFT parameters are derived exactly:

$$w = -1 - \frac{(1-\gamma)^2}{2\pi}, \quad \frac{\zeta H}{\rho} = \frac{(1-\gamma)^2}{6\pi}, \quad m_s = \frac{M_P}{N} \exp\left(-\frac{(d+1)N}{d-1}\right). \quad (24)$$

The EFT is underdamped, mass-dominated, and gradient-negligible. All 14 modes oscillate at the same frequency $\omega_0 = m_s$ and decay at the same rate $\Gamma_{\text{eff}}/2$.

The EFT layer of the five-layer loop acts as a soft projector on the residual subspace with weight $1 + c\delta$, where $c = 5/9$ is fixed by the Kernel-Diagonal Theorem (HCSM-14.a). The refined residual vacuum energy is

$$\rho_{\text{res}}^{(\text{refined})} = \frac{d+1}{2}(1+c\delta)m_\nu^4 = 2.5003 \times 10^{-47} \text{ GeV}^4, \quad (25)$$

matching observation to 0.010%.

The physical mechanism by which the residual vacuum energy becomes the observed cosmological constant is the subject of the companion sub-paper HCSM-56.a. The mechanism is pure projection onto the timelike direction of $\ker D$, with no gravitational suppression factor and no gauge mediator (HCSM-41, $\epsilon \equiv 0$).

No imported machinery appears in the EFT. The EDF Ohmic-bath layer, the 2D mass-regulated bubble, and the $\gamma_f = 1$ convention are removed. The EFT is HCSM-native.

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